Abhijatri

A Bangla-First, Offline-Ready Community Travel Platform for Bangladesh

Data Flow Diagrams

CSE 3102: Software Development Project

Team	ChoturMatra
Document	Data flow diagrams for the whole application: Level 0 (context diagram) and Level 1
Notation	Yourdon and DeMarco, as used by GeeksforGeeks
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1 Introduction

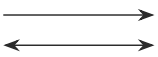
1.1 Purpose and Scope

This document models how data moves through the whole Abhijatri application: the public web app used by travellers, the staff app, the separate monitoring site and the shared API and database. It gives one Level 0 data flow diagram (the context diagram) and one Level 1 diagram that opens the single Level 0 process into seven processes. Both diagrams cover every functional requirement in the project proposal (FR01 to FR17); Section 6 shows where each requirement appears.

1.2 Notation and Rules

The diagrams use the Yourdon and DeMarco symbols that GeeksforGeeks uses in its data flow diagram articles. Table 1 shows each symbol.

Table 1. DFD Notation Used in This Document

Symbol	Name	Meaning
	**Process**	A circle with the process number and a verb phrase. It changes incoming data into outgoing data.
	**External entity**	A rectangle: a person, organisation or outside system that sends data to the system or receives data from it.
	**Data store**	Two parallel lines with the store number and name: data held at rest (a table, a file store or the device's storage).
	**Data flow**	A labelled arrow: data moving in the direction of the arrow. A two-headed arrow between a process and a store means the process reads and updates that store.

The diagrams follow these rules:

- Data flows only between an external entity and a process, between two processes, or between a process and a data store. No data flows directly between two entities, from an entity to a store, or between two stores.
- Every process has at least one input and one output, and every data store is both written and read.
- Level 0 shows the whole system as one process (numbered 0) with its external entities and no data stores.
- Level 1 splits process 0 into processes 1.0 to 7.0 and shows the data stores. It is balanced with Level 0: every flow to or from an external entity on Level 0 appears on Level 1 between the same entity and one of the seven processes (Section 5).
- Flow names are data, not actions. Each name is defined in the data dictionary (Section 4).
- To keep lines short and avoid crossings, an entity or store may be drawn more than once on Level 1. A repeated entity has a short diagonal line in its top-left corner; a repeated store has an extra vertical line at its left end.

2 Level 0 Data Flow Diagram (Context Diagram)

Figure 1 shows the whole Abhijatri application as a single process and the seven external entities that exchange data with it. Upstash Redis and Supabase Storage are not shown here: they hold the application's own data, so Level 1 shows them as data stores D8 and D7.

Table 2. External Entities

External Entity	Description
Guest	A traveller without an account. Browses, plans trips and writes guest reviews proved by a receipt.
Member	A signed-in traveller with a verified email; a Verified Member also holds the identity badge.
Staff (Moderator, Superadmin)	Moderators decide reviews, reports, appeals and verifications; the Superadmin also manages roles, destinations, prices and notification rules.
Ops Operator	The team member who reads the separate monitoring site with an ops token; not a staff role.
Supabase Auth	Outside identity service: checks passwords and MFA codes and issues signed tokens.
Cloudflare Turnstile	Outside bot check used for guest submissions.
Email Provider	Outside email service (Brevo, Resend, SMTP or Gmail) that delivers mail and reports bounces.

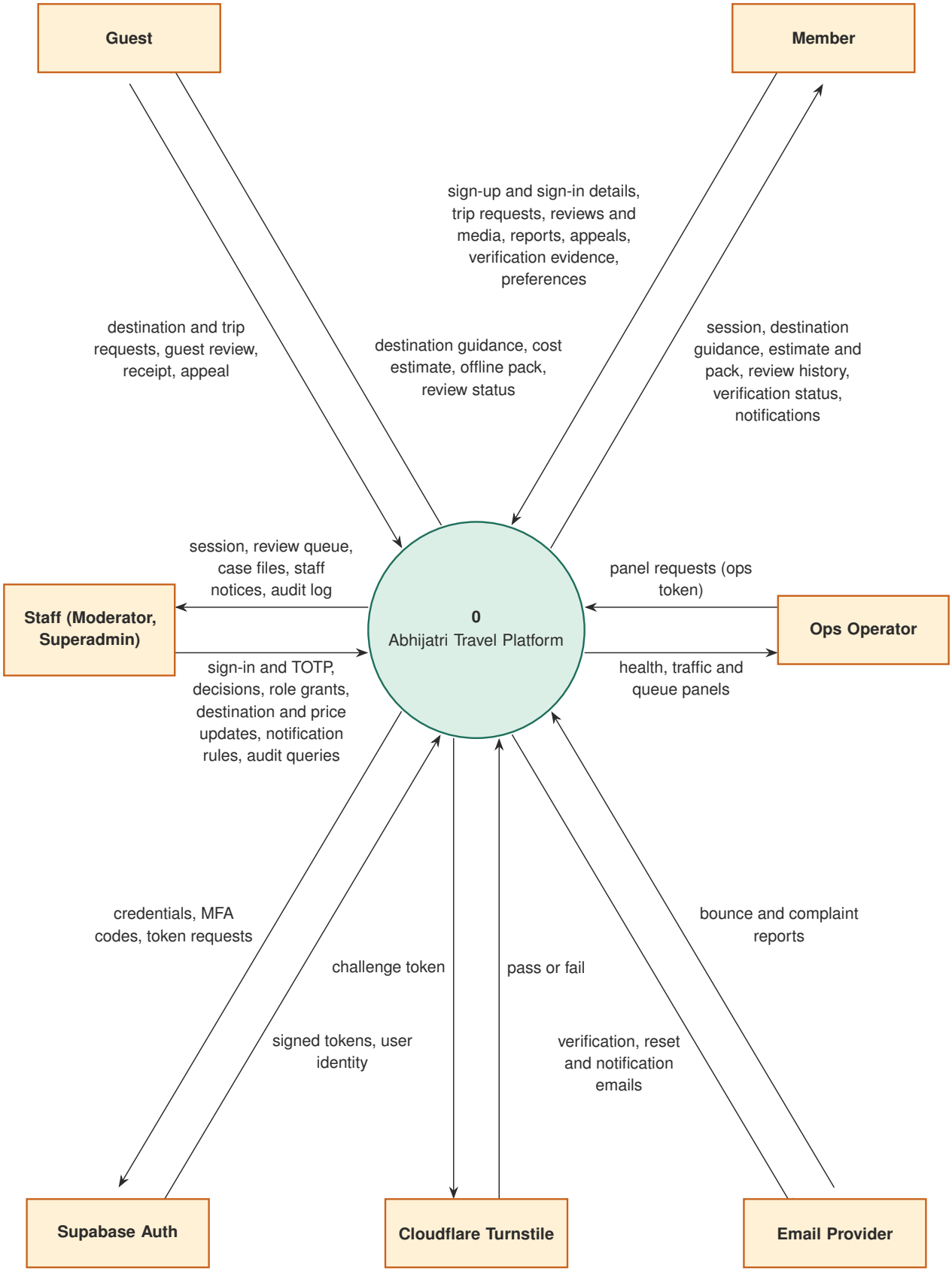


Figure 1. Level 0 Data Flow Diagram (Context Diagram) of the Abhijatri Application

3 Level 1 Data Flow Diagram

Figure 2 opens process 0 into seven processes and shows the nine data stores they share. Processes 1.0, 2.0 and 4.0 serve travellers; 3.0 handles accounts for travellers and staff; 5.0, 6.0 and 7.0 run notifications, moderation and administration. Processes pass data to one another through the stores: for example, 4.0 writes a review event to D5 and 5.0 reads it to notify the author.

Table 3. Level 1 Processes

No.	Process	What It Does	Requirements
1.0	Browse Destinations	Shows bilingual destination profiles, advisories with their freshness state, contacts, published reviews and ratings; records the visit.	FR01, FR02, FR07, FR15
2.0	Plan Trips and Keep Offline Data	Builds the itemised cost estimate from dated prices; downloads, updates and deletes offline packs; keeps trip briefs and review drafts on the device.	FR08, FR09, FR10, FR11, FR12, FR14, FR17
3.0	Manage Accounts	Registers users, verifies email, signs in members and staff (TOTP for staff), resets passwords and grants roles.	FR03, FR16
4.0	Submit and Manage Reviews	Accepts structured reviews with an idempotency key, media, edits, withdrawals, reports and appeals; checks guests with Turnstile and receipts.	FR04, FR06, FR12, FR13
5.0	Send Notifications	Turns outbox events into in-app notices and emails by rule and preference, and records bounces and complaints.	FR05, FR16
6.0	Moderate and Verify	Queues, claims and decides reviews, reports, appeals and identity verifications; updates the Community Rating in the same step.	FR05, FR06, FR07
7.0	Administer and Monitor	Lets the Superadmin update destinations, advisories and prices and read the audit log; gives the ops operator health, traffic and queue panels; removes expired data.	FR15, FR16

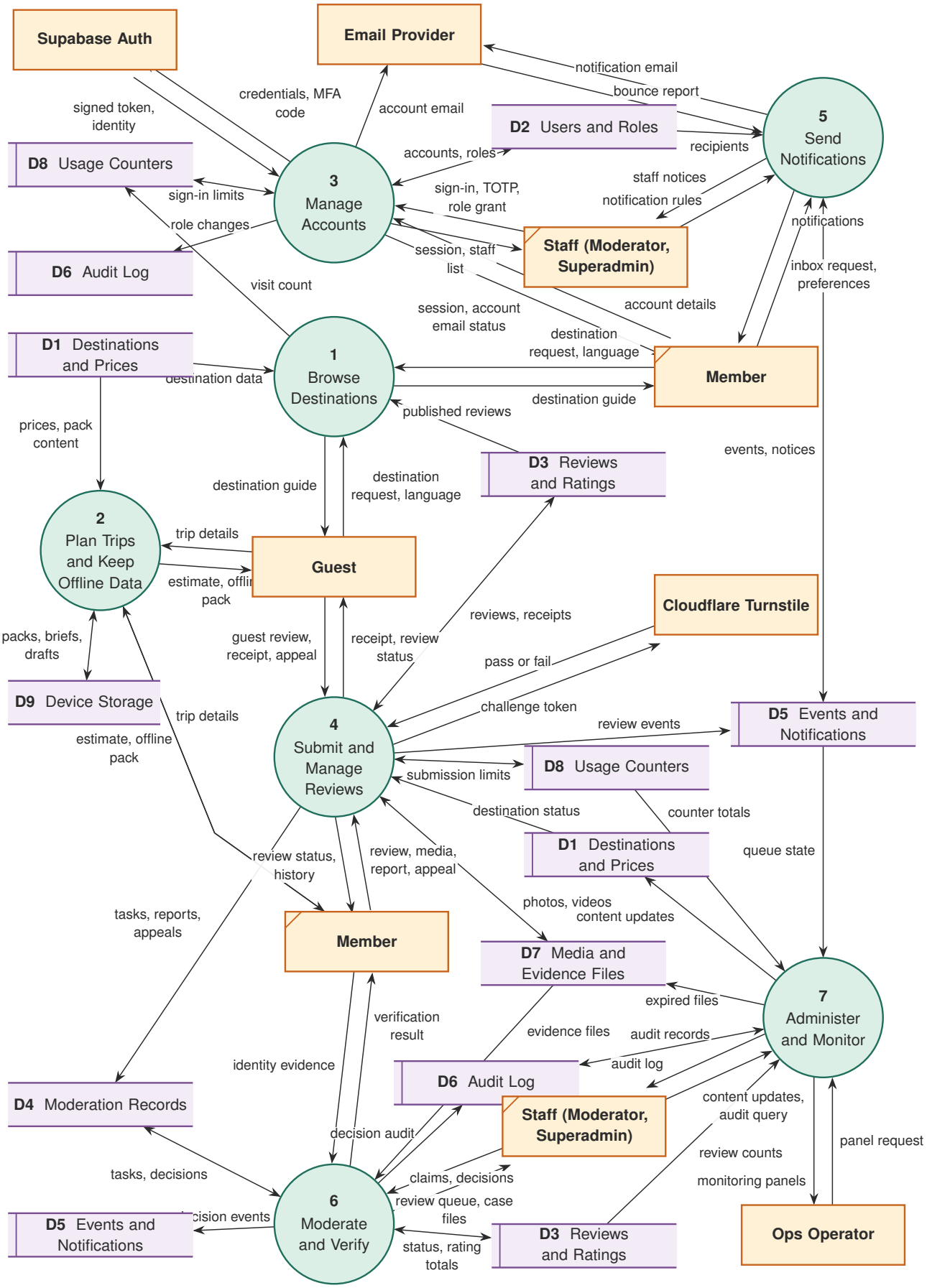


Figure 2. Level 1 Data Flow Diagram of the Abhijatri Application

Table 4. Data Stores

No.	Data Store	Data Held
D1	Destinations and Prices	Destinations, guide items, advisories with source dates, contacts and dated price records.
D2	Users and Roles	User accounts, display names, roles and visit dates (passwords stay in Supabase Auth).
D3	Reviews and Ratings	Reviews, revisions, category ratings, receipt digests, idempotency keys and destination rating totals.
D4	Moderation Records	Moderation tasks and decisions, reports, appeals, enforcement cases and verification cases.
D5	Events and Notifications	Outbox events, notifications, delivery attempts, preferences, rules and templates.
D6	Audit Log	Append-only record of every privileged action: who, what, when and why.
D7	Media and Evidence Files	Review photos and videos and verification evidence in private Supabase Storage buckets.
D8	Usage Counters	Rate-limit buckets, daily visit totals and live counters in Upstash Redis.
D9	Device Storage	IndexedDB on the traveller's device: offline packs, saved trip briefs and review drafts.

4 Data Dictionary of Level 1 Flows

Table 5 lists every data flow on the Level 1 diagram. A flow marked "read and update" is drawn with a two-headed arrow.

Table 5. Level 1 Data Flows

Data Flow	From	To	Kind
destination request, language	Guest	1.0 Browse Destinations	To or from outside
destination guide	1.0 Browse Destinations	Guest	To or from outside
trip details	Guest	2.0 Plan Trips and Keep Offline Data	To or from outside
estimate, offline pack	2.0 Plan Trips and Keep Offline Data	Guest	To or from outside
guest review, receipt, appeal	Guest	4.0 Submit and Manage Reviews	To or from outside
receipt, review status	4.0 Submit and Manage Reviews	Guest	To or from outside
destination request, language	Member	1.0 Browse Destinations	To or from outside
destination guide	1.0 Browse Destinations	Member	To or from outside
trip details	Member	2.0 Plan Trips and Keep Offline Data	To or from outside
estimate, offline pack	2.0 Plan Trips and Keep Offline Data	Member	To or from outside
account details	Member	3.0 Manage Accounts	To or from outside
session, account email status	3.0 Manage Accounts	Member	To or from outside
review, media, report, appeal	Member	4.0 Submit and Manage Reviews	To or from outside
review status, history	4.0 Submit and Manage Reviews	Member	To or from outside
inbox request, preferences	Member	5.0 Send Notifications	To or from outside
notifications	5.0 Send Notifications	Member	To or from outside
identity evidence	Member	6.0 Moderate and Verify	To or from outside
verification result	6.0 Moderate and Verify	Member	To or from outside
sign-in, TOTP, role grant	Staff (Moderator, Superadmin)	3.0 Manage Accounts	To or from outside
session, staff list	3.0 Manage Accounts	Staff (Moderator, Superadmin)	To or from outside
notification rules	Staff (Moderator, Superadmin)	5.0 Send Notifications	To or from outside

(continued)

Data Flow	From	To	Kind
staff notices	5.0 Send Notifications	Staff (Moderator, Superadmin)	To or from outside
claims, decisions	Staff (Moderator, Superadmin)	6.0 Moderate and Verify	To or from outside
review queue, case files	6.0 Moderate and Verify	Staff (Moderator, Superadmin)	To or from outside
content updates, audit query	Staff (Moderator, Superadmin)	7.0 Administer and Monitor	To or from outside
audit log	7.0 Administer and Monitor	Staff (Moderator, Superadmin)	To or from outside
panel request	Ops Operator	7.0 Administer and Monitor	To or from outside
monitoring panels	7.0 Administer and Monitor	Ops Operator	To or from outside
credentials, MFA code	3.0 Manage Accounts	Supabase Auth	To or from outside
signed token, identity	Supabase Auth	3.0 Manage Accounts	To or from outside
challenge token	4.0 Submit and Manage Reviews	Cloudflare Turnstile	To or from outside
pass or fail	Cloudflare Turnstile	4.0 Submit and Manage Reviews	To or from outside
account email	3.0 Manage Accounts	Email Provider	To or from outside
notification email	5.0 Send Notifications	Email Provider	To or from outside
bounce report	Email Provider	5.0 Send Notifications	To or from outside
destination data	D1 Destinations and Prices	1.0 Browse Destinations	One way
prices, pack content	D1 Destinations and Prices	2.0 Plan Trips and Keep Offline Data	One way
destination status	D1 Destinations and Prices	4.0 Submit and Manage Reviews	One way
content updates	7.0 Administer and Monitor	D1 Destinations and Prices	One way
accounts, roles	D2 Users and Roles	3.0 Manage Accounts	Read and update
recipients	D2 Users and Roles	5.0 Send Notifications	One way
published reviews	D3 Reviews and Ratings	1.0 Browse Destinations	One way
reviews, receipts	4.0 Submit and Manage Reviews	D3 Reviews and Ratings	Read and update
status, rating totals	6.0 Moderate and Verify	D3 Reviews and Ratings	Read and update
review counts	D3 Reviews and Ratings	7.0 Administer and Monitor	One way
tasks, reports, appeals	4.0 Submit and Manage Reviews	D4 Moderation Records	One way
tasks, decisions	6.0 Moderate and Verify	D4 Moderation Records	Read and update
review events	4.0 Submit and Manage Reviews	D5 Events and Notifications	One way
decision events	6.0 Moderate and Verify	D5 Events and Notifications	One way
events, notices	5.0 Send Notifications	D5 Events and Notifications	Read and update
queue state	D5 Events and Notifications	7.0 Administer and Monitor	One way
role changes	3.0 Manage Accounts	D6 Audit Log	One way
decision audit	6.0 Moderate and Verify	D6 Audit Log	One way
audit records	7.0 Administer and Monitor	D6 Audit Log	Read and update
photos, videos	4.0 Submit and Manage Reviews	D7 Media and Evidence Files	Read and update
evidence files	D7 Media and Evidence Files	6.0 Moderate and Verify	One way
expired files	7.0 Administer and Monitor	D7 Media and Evidence Files	One way
visit count	1.0 Browse Destinations	D8 Usage Counters	One way
sign-in limits	3.0 Manage Accounts	D8 Usage Counters	Read and update
submission limits	4.0 Submit and Manage Reviews	D8 Usage Counters	Read and update
counter totals	D8 Usage Counters	7.0 Administer and Monitor	One way
packs, briefs, drafts	2.0 Plan Trips and Keep Offline Data	D9 Device Storage	Read and update

5 Balancing Level 0 and Level 1

Each row is one flow on the Level 0 diagram and the Level 1 flows that carry the same data. Every external entity on Level 0 appears on Level 1 with the same data in the same direction, and Level 1 adds no external entity, so the two levels are balanced.

Table 6. Balancing: Level 0 Flows and the Level 1 Flows That Realise Them

Level 0 Flow	Data on Level 0	Level 1 Flows (Process)
Guest to the system	destination and trip requests, guest review, receipt, appeal	destination request, language (1.0) trip details (2.0) guest review, receipt, appeal (4.0)
The system to Guest	destination guidance, cost estimate, offline pack, review status	destination guide (1.0) estimate, offline pack (2.0) receipt, review status (4.0)
Member to the system	sign-up and sign-in details, trip requests, reviews and media, reports, appeals, verification evidence, preferences	destination request, language (1.0) trip details (2.0) account details (3.0) review, media, report, appeal (4.0) inbox request, preferences (5.0) identity evidence (6.0)
The system to Member	session, destination guidance, estimate and pack, review history, verification status, notifications	destination guide (1.0) estimate, offline pack (2.0) session, account email status (3.0) review status, history (4.0) notifications (5.0) verification result (6.0)
Staff (Moderator, Superadmin) to the system	sign-in and TOTP, decisions, role grants, destination and price updates, notification rules, audit queries	sign-in, TOTP, role grant (3.0) notification rules (5.0) claims, decisions (6.0) content updates, audit query (7.0)
The system to Staff (Moderator, Superadmin)	session, review queue, case files, staff notices, audit log	session, staff list (3.0) staff notices (5.0) review queue, case files (6.0) audit log (7.0)
Ops Operator to the system	panel requests (ops token)	panel request (7.0)
The system to Ops Operator	health, traffic and queue panels	monitoring panels (7.0)
The system to Supabase Auth	credentials, MFA codes, token requests	credentials, MFA code (3.0)
Supabase Auth to the system	signed tokens, user identity	signed token, identity (3.0)
The system to Cloudflare Turnstile	challenge token	challenge token (4.0)
Cloudflare Turnstile to the system	pass or fail	pass or fail (4.0)
The system to Email Provider	verification, reset and notification emails	account email (3.0) notification email (5.0)
Email Provider to the system	bounce and complaint reports	bounce report (5.0)

6 Requirement Coverage

Every functional requirement in the proposal is handled by at least one Level 1 process and uses the data stores shown.

Table 7. Functional Requirements and the Processes That Handle Them

ID	Requirement	Level 1 Process	Data Stores
FR01	Destination catalogue	1.0 Browse Destinations	D1
FR02	Language selection	1.0 Browse Destinations	D1
FR03	Account lifecycle	3.0 Manage Accounts	D2
FR04	Structured review	4.0 Submit and Manage Reviews	D3
FR05	Review moderation	5.0 Send Notifications, 6.0 Moderate and Verify	D3, D4, D6
FR06	Review reporting	4.0 Submit and Manage Reviews, 6.0 Moderate and Verify	D4
FR07	Destination rating	1.0 Browse Destinations, 6.0 Moderate and Verify	D3
FR08	Cost planner	2.0 Plan Trips and Keep Offline Data	D1
FR09	Pricing units	2.0 Plan Trips and Keep Offline Data	D1
FR10	Offline pack	2.0 Plan Trips and Keep Offline Data	D1, D9
FR11	Saved trip brief	2.0 Plan Trips and Keep Offline Data	D9
FR12	Offline review draft	2.0 Plan Trips and Keep Offline Data, 4.0 Submit and Manage Reviews	D9, D3
FR13	Retry-safe submission	4.0 Submit and Manage Reviews	D3
FR14	Pack update	2.0 Plan Trips and Keep Offline Data	D1, D9
FR15	Advisory freshness	1.0 Browse Destinations, 7.0 Administer and Monitor	D1
FR16	Administration	3.0 Manage Accounts, 5.0 Send Notifications, 7.0 Administer and Monitor	D1, D2, D5, D6
FR17	Storage control	2.0 Plan Trips and Keep Offline Data	D9